8. Privacy–Utility–Fairness Trade-offs

8.1 Overall Clinical Performance and Subgroup Utility

Clinical utility cannot be represented by a single test-set score. Four dimensions should be distinguished: overall discrimination, performance within clinical or demographic subgroups, calibration of predicted probabilities, and performance at the decision threshold used in practice. ROC-AUC measures ranking across thresholds, but does not show whether a protected model retains adequate sensitivity for a rare disease, acceptable specificity for screening, or reliable probabilities for triage. PR-AUC can be informative for uncommon outcomes, although it is prevalence-sensitive and should be interpreted alongside the reported outcome prevalence. The Brier score assesses overall probabilistic accuracy, whereas calibration error and reliability curves assess calibration more directly. Clinical acceptability therefore depends not only on average ranking performance, but also on whether outputs remain trustworthy and actionable in the populations in which decisions are made.

Aggregate performance can conceal unequal losses just as aggregate privacy metrics conceal highly exposed patients. A mitigation may preserve population-level ROC-AUC while reducing sensitivity for a rare diagnosis, a small demographic subgroup, or an intersectional population that contributes little weight to the average. Conversely, a modest overall loss may coexist with clinically important subgroup effects. Reporting only the protected model's mean score therefore cannot establish equitable utility. Studies should provide subgroup sample sizes, outcome prevalence, point estimates, and confidence intervals because estimates for small populations may be unstable. They should also distinguish pre-existing performance gaps from losses introduced by protection by reporting baseline utility and the absolute change for every prespecified subgroup. When many or intersectional groups are evaluated, uncertainty estimates should be accompanied by multiplicity-aware inference or appropriately specified hierarchical analyses.

Knolle et al. measured patient-linked membership vulnerability and overall diagnostic performance under stronger record-level DP in PTB-XL and EMBED [3]. Independent medical-imaging studies evaluated private training, clinical accuracy and subgroup fairness [51,52], but not the same paired patient-level MIA estimand. Together these studies support feasibility and a privacy–utility tension, without establishing how benefits and losses coincide in individual patients.

Evidence from subgroup and long-tail studies shows why equal utility loss should not be presumed. Tayebi Arasteh et al. found that useful accuracy and broadly maintained demographic fairness were feasible at intermediate DP budgets in chest radiographs and pancreatic CT, while some difficult diagnoses and subgroups still suffered larger losses [51]. Bagdasaryan et al. found larger DP-SGD accuracy reductions for underrepresented or more complex groups, including darker-skinned faces, tweets labeled as African-American English, and underrepresented classes in iNaturalist [47]. These non-clinical experiments assess accuracy rather than calibration or threshold-specific clinical harm, but show that a small aggregate loss can coexist with substantial subgroup degradation. The authors traced this pattern partly to clipping and noise acting disproportionately on uncommon or complex examples. Kulynych et al. indicate that limiting privacy disparity can involve substantial accuracy costs [7], while Truex et al. connect data skewness with minority-class membership vulnerability under different DP configurations [30]. Hu et al. further show that improving long-tail accuracy does not necessarily eliminate tail-class privacy vulnerability under DP [31]. Privacy equality and utility equality are therefore distinct objectives: improvement in one does not establish improvement in the other.

A valid comparison should evaluate baseline and protected models on identical patient-disjoint test sets and report overall and subgroup ROC-AUC or PR-AUC, sensitivity and specificity at prespecified clinical thresholds, calibration error or Brier score, worst-group performance, and confidence intervals. For each group `g`, the absolute change can be written as

$$\Delta U_g = U_g^{\text{protected}} - U_g^{\text{baseline}}.$$

For measures oriented so that higher values indicate better performance, negative values of `\Delta U_g` represent utility loss. Lower-is-better measures, such as calibration error and Brier score, should be reoriented or reported separately to avoid reversing this interpretation. Thresholds should remain

fixed from the clinical specification or be selected on an independent validation set; retuning them for every protected model can obscure changes in deployable performance. Repeated runs are necessary because private optimization adds training randomness. Comparisons should, where possible, match overall utility, privacy protection, or computational cost. Privacy-budget matching is meaningful only when the protected unit, adjacency definition, `\delta`, accountant, sampling scheme, and optimization steps are aligned [8].

8.2 Absolute Vulnerability Reduction and Disparity Reduction

Privacy mitigation has two distinct empirical objectives: reducing absolute patient vulnerability and narrowing its distribution across groups. A smaller gap can also arise if the initially less-exposed group worsens, so both levels and gaps must be reported [3,7]. Let $\widehat{V}_p^{(0)}$ and $\widehat{V}_p^{(1)}$ be the same vulnerability estimand before and after protection. It may be patient-attack AUC, a worst-record AUC summary, or another attack-specific statistic; these must never be interchanged. Both measurements require a common protected unit, membership hypothesis, attack access, score orientation and operating point, with attackers recalibrated for their respective mechanisms. Their difference is meaningful only on a fixed, interpretable difference scale; a raw likelihood ratio should normally be compared on a prespecified log scale. No posterior membership probability is implied.

On that matched vulnerability scale, the estimated absolute benefit is

$$\widehat{B}_p = \widehat{V}_p^{(0)} - \widehat{V}_p^{(1)}.$$

Positive, zero, and negative values indicate reduced exposure, no measured change, and increased exposure, respectively; uncertainty determines whether the change is credible. Benefit alone is insufficient because a large reduction may still leave a patient at an unacceptable residual vulnerability `\widehat V_p^{(1)}`. Studies should therefore report both the distribution of `\widehat B_p` and

$$\hat{\pi}^{(1)}(\tau) = \frac{1}{N}\sum_{p=1}^{N} \mathbf{1}\left[\widehat{V}_p^{(1)} \geq \tau\right],$$

for a metric-specific threshold `\tau` fixed before examining the protected model. This is the observed cohort fraction above the threshold, not a patient's membership probability; uncertainty in `\widehat V_p` should be propagated. The baseline high-risk set should likewise be fixed as

$$\mathcal{H}_0 = \left\{p : \widehat{V}_p^{(0)} \geq \tau\right\},$$

after which `\widehat B_p` within `\mathcal H_0` can be evaluated. The screening set should be selected in an independent discovery audit or by cross-fitting to limit regression to the mean in the benefit estimate. Redefining the top 1% after mitigation does not measure absolute improvement because every model has a highest percentile, although it may still describe the composition of the residual tail. Relative reduction should be reported only when the vulnerability measure has a meaningful zero and ratio-scale interpretation. It is not directly interpretable for MIA AUC, whose no-discrimination value is 0.5, or for a likelihood ratio, whose neutral value is 1. Absolute differences should therefore remain primary [3,15].

Group disparity requires separate quantities. For the audited patient cohort, let `n_g` be the number of patients in group `g` and define

$$\hat{\mu}_g^{(k)} = \frac{1}{n_g}\sum_{p:G_p=g} \widehat{V}_p^{(k)}, \qquad k \in \{0,1\},$$

and mean group benefit

$$\widehat{B}_g = \hat{\mu}_g^{(0)} - \hat{\mu}_g^{(1)}.$$

For two prespecified groups, signed and absolute mean gaps can be written as

$$\widehat{D}^{(k)}_{\text{signed}} = \hat{\mu}^{(k)}_a - \hat{\mu}^{(k)}_b, \qquad \widehat{D}^{(k)}_{\text{mean}} = \left|\hat{\mu}^{(k)}_a - \hat{\mu}^{(k)}_b\right|,$$

with disparity reduction

$$\widehat{\Delta D}_{\text{mean}} = \widehat{D}^{(0)}_{\text{mean}} - \widehat{D}^{(1)}_{\text{mean}}.$$

The signed gap identifies which group bears greater exposure, whereas the absolute gap records its magnitude. For more than two groups, prespecified pairwise contrasts, the maximum–minimum range, and worst-group vulnerability may be reported. Means can nevertheless conceal extreme exposure. Tail disparity should also be measured through

$$\hat{\pi}^{(k)}_g(\tau) = \frac{1}{n_g} \sum_{p:G_p=g} \mathbf{1}\left[\widehat{V}^{(k)}_p \geq \tau\right],$$

together with group-specific 95th and 99th percentiles. These empirical fractions classify **estimated** vulnerability; uncertainty near τ can misclassify tail membership. The tail's group composition, `P(G=g\mid V\geq\tau)`, is distinct from its within-group prevalence, `P(V\geq\tau\mid G=g)`, particularly when group sizes differ [7,30,31].

Four post-mitigation patterns illustrate why average vulnerability, tail vulnerability, disparity, and non-harm must be assessed jointly. A positive `\widehat{\Delta D}_{\mathrm{mean}}` is insufficient unless exposure decreases in the relevant groups and the proportion of patients with `\widehat B_p<0` is also reported.

Post-mitigation pattern	Overall vulnerability	Disparity	Interpretation
Similar reduction across groups	Decreases	Largely unchanged	General protection without clear gap reduction
Larger reduction for the more-exposed group	Decreases	Narrows	Preferred joint outcome
Larger reduction for the less-exposed group	Decreases	Widens	Aggregate improvement with worsening inequality
Increased vulnerability in the previously less-exposed group	May decrease, remain stable, or increase	May apparently narrow	Equality through harm; unacceptable

The clinical result in Section 8.1 lowered patient-linked vulnerability, but paired benefits and subgroup privacy gaps were not fully reported. General-benchmark studies show sample heterogeneity [15] and statistically meaningful subgroup differences under some attacks [7,30,31]. Thus a joint clinical claim of lower residual exposure **and** narrower subgroup disparity remains untested.

Table 4.1 governs matched patient and subgroup comparisons. The quantities in this section require the same vulnerability estimand and a fixed baseline set, with mechanism-matched attacker calibration and uncertainty for model repetitions and patient sampling. Attack-specific outcomes remain separate; Section 8.3 interprets paired trajectories.

8.3 Protection of High-Risk Patients and Unintended Effects

For the fixed baseline set $\mathcal{H}_0$ and benefit $\widehat{B}_p$ defined in Section 8.2, distinguish a partial decrease that remains above τ, an uncertain change, and a credible adverse increase. A conservative claim of adequate protection could require the upper confidence bound $U_p^{(1)}$ for residual vulnerability to fall below the metric-specific threshold. If a practically meaningful benefit is also required, the lower confidence bound for the paired change should exceed a prespecified $\delta_{\min}$:

$$U_p^{(1)} < \tau, \qquad L_p(\widehat{B}_p) > \delta_{\min}.$$

If `\mathcal H_0` is constructed from point estimates, it should be described as a baseline screening set. A more conservative confirmed high-risk set would require `L_p^{(0)}\geq\tau`.

Patient trajectories can be summarized using the post-mitigation set

$$\mathcal{H}_1 = \left\{ p : \widehat{V}_p^{(1)} \geq \tau \right\}.$$

The resulting transitions are more informative than redefining the top 1% after protection, because percentile-based lists necessarily contain the same proportion of patients and may show tail replacement even when absolute exposure has declined.

Transition state	Definition	Interpretation
Persistently high risk	`\mathcal H_0\cap\mathcal H_1`	Remained above the threshold
Protected	`\mathcal H_0\setminus\mathcal H_1`	Left the high-risk region
Newly high risk	`\mathcal H_1\setminus\mathcal H_0`	Entered the region after mitigation
Persistently lower risk	`(\mathcal H_0\cup\mathcal H_1)^c`	Remained below the threshold

These four transitions describe point-estimate trajectories. Patients whose uncertainty intervals overlap `\tau` should be reported separately as indeterminate rather than assigned definitively to the protected or newly high-risk categories.

The clinical DP analysis in Section 8.1 did not establish that the same patients left the tail under each level. On general benchmarks, binary low-FPR vulnerable sets varied more across training runs than continuous rankings [26]. Intervals and repeated transitions are needed before calling a newly listed patient newly exposed.

Intervention can also change which patients are most exposed. The Privacy Onion Effect shows that memorization is relative to the surrounding training distribution: removing highly vulnerable examples may reveal another layer whose exposure was previously obscured [35]. It demonstrates that vulnerability transfer is possible when intervention changes the training distribution, not that every mitigation creates new high-risk patients. Vulnerability reduction means exposure falls without important new vulnerabilities; vulnerability redistribution means targeted patients improve while others worsen; tail replacement means absolute vulnerabilities decline but different patients occupy the highest ranks; and apparent transfer arises from uncertainty, calibration changes, or incompatible audits. Genuine redistribution requires paired increases for other patients on a comparable scale; altered ranks or low Jaccard overlap alone are insufficient.

Vulnerability-guided protection may create further harms. Downweighting rare patients could reduce their influence while degrading the model's ability to learn rare conditions; this potential utility harm requires direct evaluation. Small groups may additionally receive less stable vulnerability estimates and apparently unequal benefits [7]. Individualized DP-SGD demonstrates feasible non-uniform privacy budgets, but uses prespecified assignments rather than audited clinical vulnerability [45]. REPS more directly uses estimated outlierness to guide differentially private synthetic-data generation [46]. However, it is record-level, concerns synthetic tabular data, depends on its risk scorer, and has not

validated protection for multi-record patients in diagnostic models. Risk labels and personalized budgets may themselves be sensitive metadata: identifying someone as high risk could reveal that their data are unusual or associated with a rare condition. Unstable estimates may also produce inconsistent protection across retraining cycles.

Report definitive and indeterminate transitions, paired $\widehat{B}_p$ with intervals, meaningful increases among other patients, and subgroup clinical utility. Rank turnover or low set overlap alone is not evidence of transfer. Compare with a uniform intervention under prespecified matched constraints; use the complete Table 4.1 protocol for attack adaptation and robustness.

8.4 Computational Cost and Practical Feasibility

Practical feasibility is not determined by GPU time alone. It includes repeated training, reference construction, calibration, queries, statistical precision, private optimization, storage, patient linkage, monitoring, governance of identifiers and scores, and expert review. The question is whether an audit can produce reliable and actionable patient-level conclusions within computational, data, governance, and time constraints.

Reliable patient-level auditing requires repeated observations under patient inclusion and exclusion. A first-order approximation of its cost is

$$C_{\text{audit}} \approx M_T C_T + M_R C_R + C_{\text{scoring}}(M, N, A) + C_{\text{calibration}},$$

Here, `M_T` and `M_R` count target and reference models, `C_T` and `C_R` are their training costs, `N` is the number of evaluated units, and `A` is the number of attacks [2,3]. Knolle et al. trained 200 target models and reused them across patients, amortizing rather than repeating this cost per patient [3]. Leave-one-patient-out retraining scales poorly, whereas shared subsampling reuses models. Offline or one-shot attacks are cheaper but implement different adversarial experiments and may not provide enough counterfactual observations for patient-specific uncertainty or low-FPR estimation [2].

Reducing the number of models is not merely an engineering compromise. Too few models produce wide intervals, unstable threshold classifications, unreliable tails, and insufficient resolution at low FPR. Jälkö et al. warn that pooled efficient evaluations can hide per-sample FPR heterogeneity and introduce finite-population bias; their study remains a preprint [14]. Score storage can scale with the product of evaluated units and models, while checkpoints add model-size-dependent costs. Sufficient statistics can reduce storage if reproducibility is preserved. Influence or difficulty measures may prioritize detailed audits, but their efficiency does not make them privacy estimates.

Protection adds a second cycle. DP-SGD requires contribution clipping, noise, accounting, and tuning; its overhead depends on architecture, batching, implementation, and hardware [8,52]. Patient-level DP additionally samples patients, aggregates their records, handles unequal contribution sizes, and accounts under patient adjacency. Individualized DP-SGD manages heterogeneous sampling or clipping parameters [45]. Tuning may itself require accounting or separate validation data, and protected training must be followed by independent privacy and utility re-auditing.

Clinical institutions face constraints absent from benchmarks. Medical histopathology and EHR studies illustrate the feasibility and practical difficulty of private distributed training, without providing patient-specific MIA benefit estimates [54,55]. Auditing requires stable identifiers to link records belonging to one patient while preventing patient leakage across target, test, and reference sets. Cross-institution linkage may be legally or technically restricted, and differences in devices, acquisition protocols, preprocessing, and population distributions can compromise reference-model validity. Rare and intersectional groups may be too small for precise estimation. Vulnerability estimates must also be secured because they may reveal unusual or sensitive clinical characteristics. An audit may become obsolete after changes to the dataset, architecture, preprocessing, or privacy mechanism. Re-audit triggers should therefore be specified prospectively rather than treating one result as permanent certification [3].

A tiered workflow can allocate resources more realistically, but aggregate screening must characterize system-level leakage rather than act as a gate that determines whether patient-level auditing

occurs. A prespecified sentinel sample should undergo individual auditing even when aggregate results appear reassuring.

Stage	Purpose	Relative cost	Main limitation
System-level characterization	Measure aggregate leakage and model behavior	Low	Cannot rule out an extreme-risk tail
Sentinel patient-level audit	Audit a prespecified stratified sample, including rare and multi-record patients	Moderate–high	May still miss very rare vulnerabilities
Expanded patient-specific audit	Extend auditing to flagged strata and random controls	High	Requires repeated models and calibration
Targeted causal validation	Apply leave-one-patient-out checks to selected cases	Very high	Cannot readily cover the full population
Protected-model re-audit	Verify benefit and detect redistribution	High	Must remain independent of intervention selection

The saving from tiering depends on the design. Shared target models amortize training across patients, whereas target-specific references or leave-one-patient-out retraining scale with the number audited. Tiering is therefore a prioritization strategy, not proof that unscreened patients are safe. Screening sensitivity must be validated against detailed audits in a prespecified stratified and random sample.

Feasibility reports should include models, IN/OUT observations, patient and record counts, queries, time, storage and the FPR precision actually supported. Tiered audits prioritize resources, but their sentinel sample must be validated against detailed random and stratified audits. These constraints inform Section 9.